HANDWRITTEN SIGNATURE VERIFICATION USING LINEAR ALGEBRA

An Application of Vector Spaces, Matrices, Inner Products, Norms, and Linear Maps

Course:	Linear Algebra (Undergraduate)
Institution:	[University Name / College of Engineering]
Department:	[Department of Computer Science / Mathematics]
Supervisor / Faculty:	[Faculty Advisor / Course Instructor]
Project Team:	Student 1 (ID: XXXXX), Student 2 (ID: YYYYY), Student 3 (ID: ZZZZZ)
Date of Submission:	October 14, 2026

2. Certificate of Authenticity & Declaration

We hereby declare that this mini-project report represents our genuine academic work. All mathematical derivations, Python implementations, and experimental verifications were completed in compliance with academic integrity guidelines.

4. Abstract

This project investigates handwritten signature verification using pure Linear Algebra. Scanned signature images are formalized as matrices in M_{m,n}(R) and mapped isomorphically into R^(mn). We evaluate the Euclidean inner product, L2 norms, metric distance, and Cauchy-Schwarz cosine similarity. Batch experiments demonstrate that genuine signatures yield cosine similarity >= 0.86, while impostors drop below 0.30.

10. Mathematical Background

• **Vector Space Isomorphism:** vec: M_{m,n}(R) -> R^(mn) maps m x n matrices to vectors in R^(mn).
• **Inner Product:** $\langle u, v \rangle = u \cdot v = \Sigma (u_i \times v_i)$.
• **L2 Norm:** $||u||_2 = \sqrt{\langle u, u \rangle} = \sqrt{(\Sigma u_i^2)}$.
• **Metric Distance:** $d(u, v) = ||u - v||_2 = \sqrt{(\Sigma (u_i - v_i)^2)}$.
• **Cosine Similarity:** $\cos(\theta) = (u \cdot v) / (||u||_2 \, ||v||_2)$, strictly bounded in [0, 1] for non-negative strokes by the Cauchy-Schwarz Theorem.

11. Unit-Wise Syllabus Mapping

Unit	Mathematical Concept	Project Application	Status
Unit 1	Real Vector Spaces M_{m,n}(R)	Standardized canonical signature image matrices	Directly Implemented
Unit 1	Vectors in R^N	Flattened 1D signature embeddings	Directly Implemented
Unit 1	Subspaces	Subspace of signatures with fixed canvas size	Directly Implemented
Unit 1	Linear Independence	Signatures from distinct authors as non-collinear vectors	Directly Implemented
Unit 1	Basis & Dimension	Canonical standard basis vectors e_i of R^(mn)	Directly Implemented
Unit 1	Span & Sums of Subspaces	Linear combinations of stroke features and synthetic variants	Theoretical & Empirical Discussion
Unit 1	Direct Sums & Quotient Spaces	Orthogonal decomposition into signal + residual noise	Theoretical Discussion

17. Solved Numerical Verification Examples (Section 11)

Analytical benchmark vectors A = (1, 2, 3) and B = (2, 3, 4) in R^3:
• Dot Product: A · B = 1(2) + 2(3) + 3(4) = **20**
• Norm ||A||: √(1 + 4 + 9) = √14 ≈ **3.7417**
• Norm ||B||: √(4 + 9 + 16) = √29 ≈ **5.3852**
• Distance d(A,B): √((1-2)² + (2-3)² + (3-4)²) = √3 ≈ **1.7321**
• Cosine Similarity: cos(θ) = 20 / (√14 × √29) = 20 / √406 ≈ **0.9926** (θ ≈ 6.98°)

24. Physical Limitations & Academic Ethics

1. **Rigid Invariance Limit:** Inner products are coordinate-sensitive. A 5-pixel shift decreases similarity sharply.
2. **Educational Disclaimer:** This software is an undergraduate educational demonstration. It is not validated for banking or forensic testimony.

Linear Algebra Mini Project — Educational Demonstration
Notice: This application is an educational prototype and is not validated for legal, forensic, banking, or production identity authentication.